

RESEARCH ARTICLE SUMMARY

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Molecular cascades and cell type-specific signatures in ASD revealed by single-cell genomics

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INTRODUCTION: Historically, psychiatric disorders have been distinguished from neurological disorders by the absence of the associated histological pathology observed in neurological conditions. But, over the past 15 years, epigenetic and transcriptional profiling of postmortem brain samples from multiple psychiatric conditions, including autism spectrum disorder (ASD), have revealed robust underlying molecular differences. In ASD, this reflects up-regulation of immune signaling genes, down-regulation of neuronal markers and synaptic genes, and a blunting of gene expression sig-

natures of cortical regional identity. How a genetically complex condition such as ASD converges on shared transcriptional alterations remains a mystery. This gap is further amplified by the lack of biological insights into the differences in laminar and cell type-specific pathways affected in ASD and the underlying gene regulatory mechanisms.

RATIONALE: We reasoned that deep molecular profiling at the single-cell level would inform our understanding of changes in cortical cell types and circuits and, when integrated with

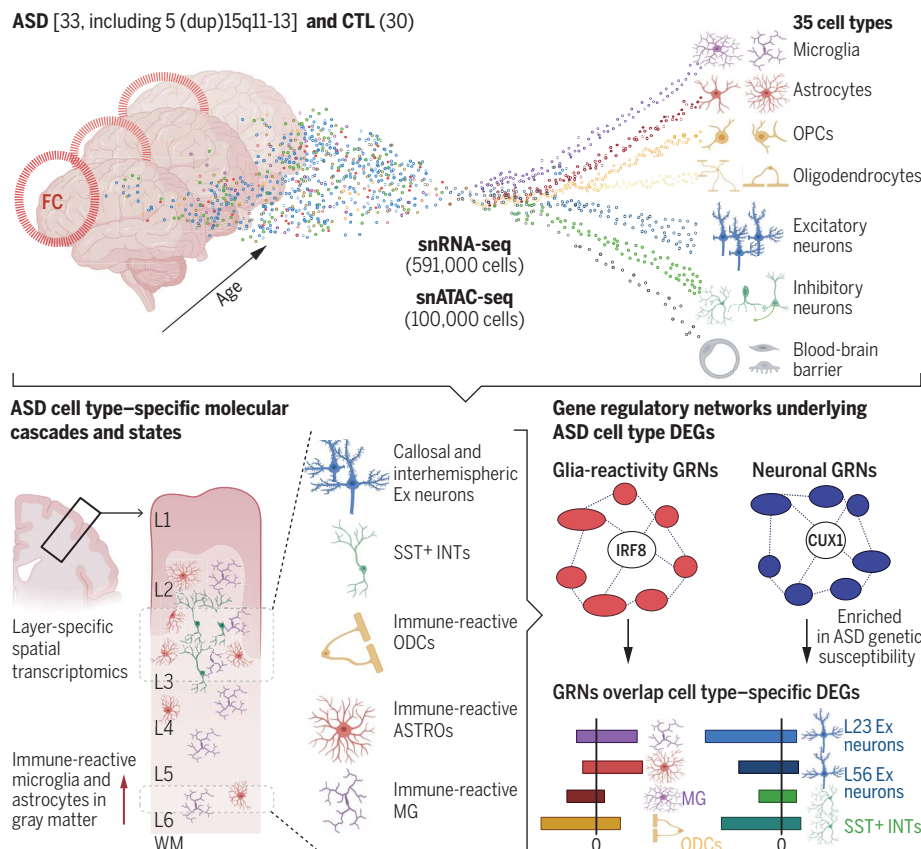
genetic risk factors, enable identification of candidate drivers of pathways altered in ASD. Further, knowledge of these pathways at a cellular level would inform mechanistically driven therapeutic development.

RESULTS: We conducted single-nucleus RNA sequencing (snRNA-seq) and single-nucleus assay for transposase-accessible chromatin with sequencing (snATAC-seq) in a large ASD and control (CTL) cohort, as a core component of the PsychENCODE Consortium (<https://www.psychencode.org/>), to identify cell type-specific changes and the cellular regulatory networks perturbed by genetic risk factors. Our approach (on average >10,000 cells per individual and >1860 genes per cell) enabled identification of all 26 major cortical cell types, validated with published cortical cell atlases. We also identified cell states, primarily activated forms of microglia (MG), oligodendrocytes (ODCs), astrocytes (ASTROs), and blood-brain barrier cells, observed predominantly in ASD. Changes in cell composition in ASD were subtle, involving increases in activated microglia and astrocyte states, which are very rarely observed in controls.

In contrast to the minor changes in cell composition, the changes observed in gene expression in ASD were substantial: 2166 down-regulated and 1319 up-regulated genes across 35 cell types, most of which were cell type specific. Through integration of snRNA-seq, snATAC-seq, and spatial transcriptomics, we identified regulatory networks driving cell type-specific transcriptional changes and their location within cortical laminae. These analyses demonstrate concentration of activated microglia, astrocytes, and somatostatin (SST) interneurons in superficial cortical laminae, in conjunction with profound down-regulation of synaptic gene expression and up-regulation of stress-response and proinflammatory pathways in inter- and intrahemispheric projection neurons. A large proportion of these changes could be ascribed to specific transcriptional drivers, and both the drivers and their targets were enriched in genes harboring common and rare genetic risk for ASD.

CONCLUSION: These analyses refine our knowledge of cellular and circuit alterations in the brain in ASD. By identifying and validating transcriptional drivers enriched in rare and common genetic risk variants, we have discovered a link between autism genetic susceptibility and molecular and cellular circuits and pathways, providing a roadmap for understanding cellular interactions and therapeutic development in ASD. ■

ASD [33, including 5 (dup)15q11-13] and CTL (30)



Single-cell genomics reveals cell type-specific and laminar changes in ASD. These changes prominently affect layers 2 and 3 interhemispheric and callosal-projecting excitatory (Ex) neurons, superficial SST interneurons, and reactive glial states in the frontal cortex (FC). By defining gene regulatory networks (GRNs; red, up-regulated; blue, down-regulated) and integrating them with ASD genetic risk variants, we discerned candidate drivers of the transcriptional changes and genetic susceptibility acting in specific cell types. OPCs, oligodendrocyte progenitor cells; INTs, inhibitory neurons; WM, white matter; DEGs, differentially expressed genes. [Created with Biorender.com]

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RESEARCH ARTICLE

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Molecular cascades and cell type-specific signatures in ASD revealed by single-cell genomics

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Genomic profiling in postmortem brain from autistic individuals has consistently revealed convergent molecular changes. What drives these changes and how they relate to genetic susceptibility in this complex condition are not well understood. We performed deep single-nucleus RNA sequencing (snRNA-seq) to examine cell composition and transcriptomics, identifying dysregulation of cell type-specific gene regulatory networks (GRNs) in autism spectrum disorder (ASD), which we corroborated using single-nucleus assay for transposase-accessible chromatin with sequencing (snATAC-seq) and spatial transcriptomics. Transcriptomic changes were primarily cell type specific, involving multiple cell types, most prominently interhemispheric and callosal-projecting neurons, interneurons within superficial laminae, and distinct glial reactive states involving oligodendrocytes, microglia, and astrocytes. Autism-associated GRN drivers and their targets were enriched in rare and common genetic risk variants, connecting autism genetic susceptibility and cellular and circuit alterations in the human brain.

Psychiatric disorders are defined primarily by behavioral and cognitive characteristics and are classically distinguished from neurological disorders by the absence of the associated visible histological pathology observed in neurological conditions. However, a growing body of evidence based on genomic profiling reveals consistent molecular differences in brain tissue from specific neuropsychiatric conditions compared with brain tissue from neurotypical individuals (1–5). In autism spectrum disorder (ASD), robust transcriptomic and epigenetic alterations in the cerebral cortex have been documented over the past decade, delineating a reproducible pattern of molecular differences (5–13). This robust molecular signature obtained primarily from transcriptomic profiling of bulk cortical tissue has identified convergent biological pathways in the ASD brain, which is characterized by an up-regulation of immune signaling genes, down-regulation of specific neuronal markers and synaptic genes, and an attenuation of the typical patterns of gene expression associated with cortical regional identity (6–8, 12–14).

These genomic data represent an essential lens through which to understand the cellular

and physiological changes occurring in the brains of autistic individuals and to describe potential causal mechanisms through their integration with genetic risk variants (1, 4, 5). However, small sample sizes, in the case of single-cell analysis (13), or profiling restricted to bulk tissue have limited biological insights as to the differences in laminar, circuit-level, and cell type-specific pathways affected in ASD, as well as their underlying gene regulatory mechanisms. To address these limitations, we leveraged advances in single-cell analyses to profile the largest ASD cohort to date, consisting of 63 cases and controls, including five cases of a monogenic cause of ASD, 15q11-13 duplication [(dup)15q11-13]. This resource, generated as a core component of the PsychENCODE Consortium (1, 2, 4, 5, 8), enabled our identification of the major cortical cell types affected in the ASD cortex, including alterations in glia cell composition and cell type-specific transcriptional cascades linking genes to function. Through integration of single-nucleus RNA sequencing (snRNA-seq), single-nucleus assay for transposase-accessible chromatin with sequencing (snATAC-seq), and layer-specific spatial transcriptomics, we describe and validate regulatory networks driving cell type-specific transcriptional changes and their location within cortical laminae, providing a deeper understanding of the biological mechanisms that underlie ASD.

Results

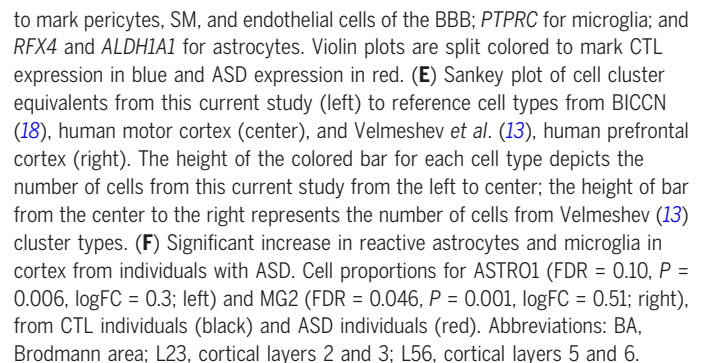
Single-cell genomic analysis of human ASD and control cortex

We generated snRNA-seq from the human frontal cortex (Brodmann areas 9 and 4/6)

from 33 ASD and 30 control subjects (Fig. 1A and table S1A), profiling, on average, >10,000 cells per individual (fig. S1, A and B). We included five individuals with a monogenic form of ASD, 15q duplication syndrome [(dup)15q11-13; table S1A], where transcriptomic changes in bulk tissue mirror those observed from cases of idiopathic ASD (7, 8). Individual samples were taken from patients ranging in age from 2 to 60 years (mean age: 20) and were selected to be matched by age, sex, and cause of death balanced across cases and controls (fig. S1, B to D, and table S1A). We isolated >800,000 nuclei and performed snRNA-seq using the 10x Genomics platform (Materials and methods). After quality control and doublet removal, we analyzed high-quality transcriptomes from 591,043 nuclei representing 306,347 cells from ASD subjects and 284,696 cells from control subjects (average read depth of 6.6×10^4 per cell) with a mean of 1860 genes and 4164 RNA molecules detected per cell (fig. S1B), which provided an increase in power over previous work (13) (table S1B). We simultaneously isolated nuclei to generate chromatin conformation states using snATAC-seq in 22 cases and controls to validate predicted cell type-specific regulatory networks through identification of enriched transcription factor (TF) binding sites with open chromatin and TF footprinting analyses (Fig. 1A).

We performed unsupervised clustering of transcriptional signatures from both ASD cases and controls while considering biological and technical sources of variation (methods). Visualization of single-nuclei transcriptomes grouped nuclei into 35 clusters that were not driven by batch, Brodmann area, or individual samples (Fig. 1, B and C, and fig. S1, A, C, and D). We annotated the clusters into seven major classes of cortical cell types—neurons, specifically excitatory neurons (Ex neurons) and inhibitory neurons (INTs); glial cells, including microglia (MG), astrocytes (ASTRO), oligodendrocyte progenitor cells (OPCs), and oligodendrocytes (ODCs); and blood-brain barrier (BBB) cells—on the basis of previously established cell type-specific canonical gene sets from both human and mouse datasets (15–19), revealing the expected neuronal-to-glial ratio within human cortex (18) (Fig. 1D and fig. S1E). Next, we annotated distinct subgroups within each major class by aligning our cell clusters with reference data from the Brain Initiative Cell Census Network (BICCN) (18) and the Allen Brain Atlas (19), as well as a previous snRNA-seq dataset from individuals with ASD and controls (13) and using cluster-specific differential marker genes (table S1C; methods). Each of our 35 clusters corresponded with the cell profiles previously defined (Fig. 1E and table S1D), representing twice the number of cell clusters identified in the previous, smaller ASD single-cell analysis (13). Additionally, given the

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larger number of cells and cohort of individuals profiled here, we defined a number of cell types and states not previously described in the human brain atlas (18, 19) (Fig. 1E and table S1, B and D), including ASD-specific states, which mostly represent BBB cells and reactive glial states.

Within non-neuronal populations, BBB subtypes separated according to specific markers into endothelial cells, pericytes, and smooth muscle (SM) cell types (20) (Fig. 1, D and E, and fig. S2A). Astrocytes separated into three clusters by expression of specific markers annotating two gray matter or protoplasmic astrocyte clusters, one homeostatic (ASTRO2) and one reactive (ASTRO1), and one white matter astrocyte cluster (ASTRO3) (21) (fig. S2B). We also generated a small spatial transcriptomics dataset (Fig. 1A and fig. S2C; methods) with sufficient resolution to map specific cellular signatures onto their locations relative to cortical laminae and white matter. To map these cluster-specific signatures onto the cortical layers, we filtered cluster marker gene sets from the snRNA-seq data into sets of differentially expressed genes for each cell type (methods and table S1C). Using this strategy, we confirmed the known patterns of cell position in the cortex. White matter or fibrous astrocytes were located in white matter and layer 1, whereas reactive protoplasmic astrocytes were found in the gray matter layers (22) (fig. S2D). ODC cluster expression profiles were separated by their maturity and myelination capabilities (ODC1 being the most mature, and ODC3 the least mature) (23) (fig. S2E). OPC clusters separated according to maturation of myelinating properties and subtype, with OPCS2 expressing the most-mature myelination profile, and OPCS1 being immature cells, whereas OPCS3 were delineated by *BCAS1* (brain enriched myelin associated protein 1) and *GPR17* (G protein-coupled receptor 17), which indicates that they are a specific type of OPC that supports synaptic function (24, 25) (fig. S2F). MG subtypes were mainly split into two states: one homeostatic (MG1) and the other reactive (MG2 & BBB_MG3), differentiated by the expression of known markers (26) (fig. S2G). Homeostatic MG1 was found primarily within white matter and was much less abundant in gray matter. MG2 distribution was very similar to distribution of homeostatic MG1, with greater abundance in layer 1 (fig. S2H, left and middle panels).

Ex neuron subtypes were separated by marker genes associated with known superficial-to-deep cortical distribution and axonal projection trajectories (27, 28) (fig. S2I). For Ex neurons, we found that, in all cases, the cluster expression profiles predict their locations within the spatial transcriptomics data—with layer 2/3 Ex neurons' marker gene expression located in the superficial layers, and layer 5/6 Ex neurons' profiles observed most highly in deeper layers

(fig. S2J). INT subtypes were annotated on the basis of canonical marker genes and differential gene expression associated with INT ontologies (27–30). We defined one somatostatin (SST+) cluster, two parvalbumin (PV+) clusters, one vasointestinal peptide cluster (VIP+) that also contained calretinin cells (CR+), one reelin (RE+) cluster, and another cluster that consisted of tyrosine-protein kinase KIT (KIT+) cells and synaptic vesicle glycoprotein 2C (SV2C+) cells (fig. S3A). Mapping the cluster-specific profiles shows that these INT subtypes were distributed evenly across laminae in the spatial data, with the exception of the caudal ganglionic eminence (CGE)-derived cluster signature, which was less abundant in layer 4, matching its known cortical distributions in humans (31) (fig. S3B).

Cell composition changes in ASD

We next performed cell composition analysis to determine whether any cell types were reduced or more abundant in the ASD cortex compared with controls (see methods). In agreement with previous analysis (13, 32, 33), we detected a small but significant increase in reactive forms of microglia ($P = 0.04$) and a similar trend in astrocytes ($P = 0.10$) (Fig. 1F and table S2, A and B). The laminar distribution of the reactive MG2 cluster signature in ASD from the spatial transcriptomics data demonstrated an increase in MG2 within gray matter laminae in ASD compared with control (Fig. 1F and fig. S2H, center panel). We observed no other significant changes in glial or neuronal cell type abundance, so we next assessed cell type-specific transcriptional changes.

Overview of the cell type-specific transcriptional changes in ASD

We detected widespread transcriptional changes encompassing most cell types; we identified 3485 significant differentially expressed genes (DEGs) comprising 2166 down-regulated and 1319 up-regulated genes across 35 cell types [false discovery rate (FDR) < 0.05; log fold change ($\log_{2}FC$) $\geq \pm 0.2$; pseudobulk approach, see methods; Fig. 2A], representing 2810 different genes, of which only 675 (24%) were differentially expressed across more than one cell type. This is more than five times the number of DEGs compared with the previous single-cell study (13). We also confirmed that subjects with 15q dup syndrome (fig. S4, A and B) harbored a differential expression signature similar to, but of higher magnitude than, idiopathic cases of ASD, with Spearman correlations (R^2) of DEGs across each cell type ranging from 0.76 to 0.97 ($P = 0.0$ to 2.3×10^{-20} ; FDR < 0.01 threshold; fig. S4, C and D). We also observed high concordance between these transcriptional changes and a previous ASD dataset profiled by bulk RNA-seq (7) ($R^2 = 0.86$, $P = 0$, Spearman correlation; fig. S4E). Given that approximately half of our samples were not part of any previous bulk tissue or

single-cell studies, in part, this analysis represents an independent replication. Notably, ASD susceptibility genes were enriched in DEGs overall: Out of 2810 DEGs, 333 (12%) genes were known syndromic and tier 1 or 2 ASD vulnerability genes from the SFARI Gene database (34–40) [odds ratio (OR) = 4.1, $P = 4.9 \times 10^{-152}$, hypergeometric test], and 71 genes were in the syndromic or tier 1 SFARI categories (OR = 1.4, $P = 7.7 \times 10^{-29}$ hypergeometric test; methods).

Most changes were cell type specific, and DEGs were not evenly distributed across cell types, confirming previous observations (13). Superficial Ex neurons (Ext_6_L23) and reactive glial cell types (ASTRO, MG2, and ODC1) had the highest number of DEGs and the highest number of changes in gene expression that were not detected in other cell types (Fig. 2A). We also observed that the effect of the 15q11-13 duplication displayed some distinct changes, including marked cell type-specific effects on DEGs within the duplication region, both within neuronal subtypes and between neurons and glia (fig. S4F), illustrating the cell type-specific effects of gene dosage. These data suggest that the cell type-specific regulatory architecture is largely intact in the (dup)15q11-13 brain. Neuronal clusters expressed the highest number of genes within the duplication region and showed the highest number of genes up-regulated (fig. S4F, bottom), whereas among glia, ASD-associated activated states (OPCS1, ASTRO1, and MG2) exhibited the highest proportion of up-regulated genes within this region (fig. S4F, top).

Cell type-specific state changes in ASD

Next, we explored the pathways and mechanisms of cell type-specific transcriptional changes. The most DEGs were detected in superficial Ex neurons (Ext_6_L23); cortical-projecting deep Ex neurons (EX_4_L56); SST+ INTs (INT_2_SST); two reactive astrocyte clusters, one protoplasmic and one fibrous (ASTRO1/ASTRO3); and reactive MG2, OPCS1, and ODC1 (Fig. 2A) (41). We also used these single-cell expression data to annotate coexpression modules identified in previous bulk gene expression data, identifying 23 modules that were previously associated with ASD but that we now can show exhibit prominent cell type specificity (table S3).

Neuronal changes

Most DEGs in neurons were down-regulated and enriched in Gene Ontology (GO) annotations that converged on synaptic components, both postsynaptic and presynaptic, as exemplified by changes in L23 Ex neurons, L56 Ex neurons, and SST+ INTs (Fig. 2, B and C; table S4, A to C; and fig. S5, A and B; methods). For superficial Ex neurons (EXT_6_L23), the most significantly down-regulated pathways were related to dendrite or cell projection (axon) organization (including *NETO2*, *GRIPI*, *DLGAP3*, *ARC*, and *ATAD1*), vesicle regulation (including

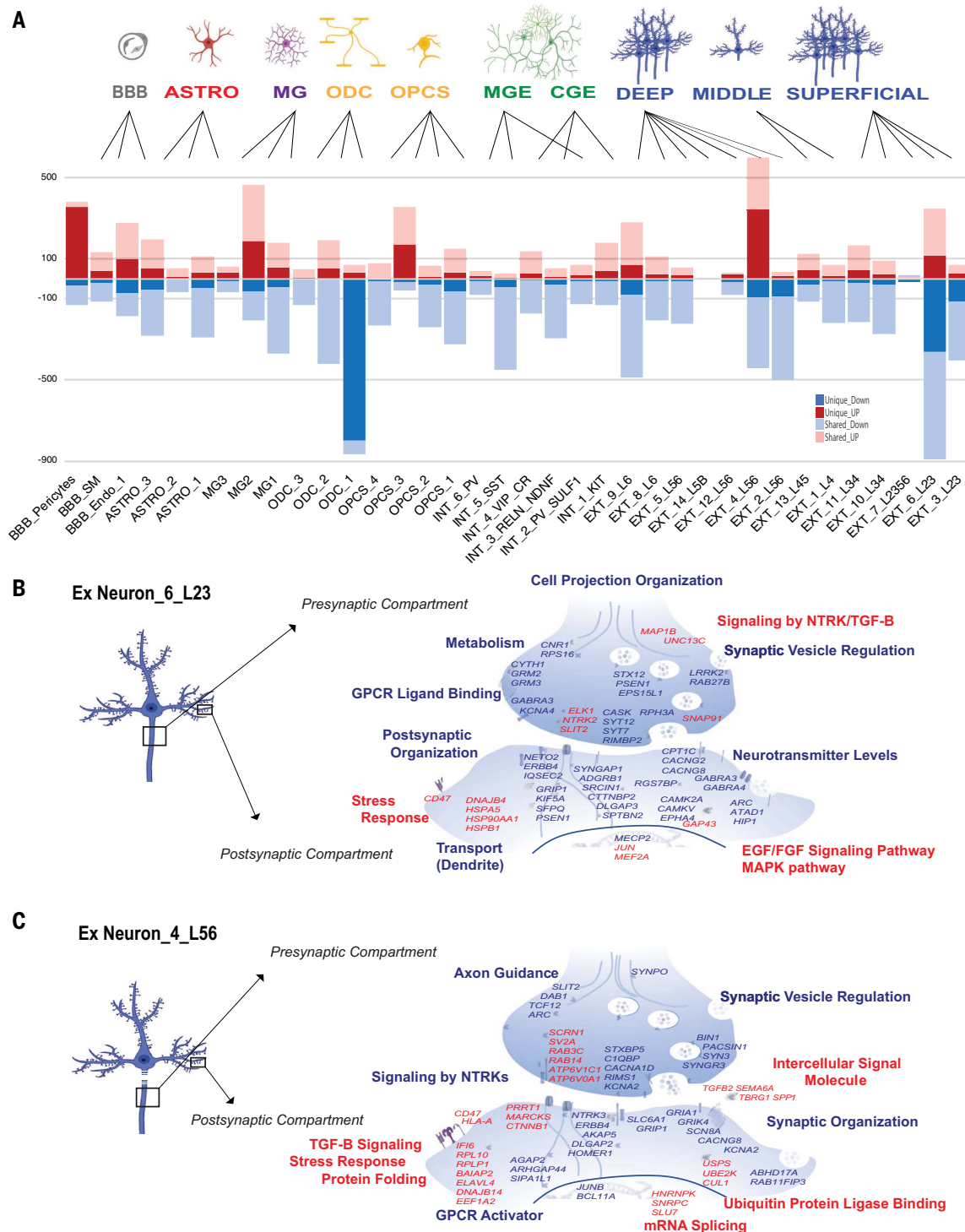


Fig. 2. Cell type-specific transcriptional changes. (A) Cell type-specific count of up-regulated genes (red) and down-regulated genes (blue). Ex neurons are labeled by predicted layer identity (blue); INTs are marked by whether they are medial ganglion eminence (MGE)-derived (PV+ and SST+ INTs) or caudal ganglionic eminence (CGE)-derived (VIP+, CR+, RELN/NDNF+, KIT/SV2C+) (green); OPCs and ODCs are labeled in yellow; MG are labeled in purple; ASTROs are labeled in red; and BBB cells are labeled in gray. The numbers of shared genes (dark-red or dark-blue bar) and the numbers of unique genes (light-red or light-blue bar) vary by cell type. (B) Schematic of Ex neuron (blue) cell soma, dendrites, and initial axonal projection. Black squares indicate putative location of

the enlarged example of the presynaptic axonal compartment (top) and enlarged example of postsynaptic dendritic compartment (bottom). The most significant up-regulated GO terms and pathways (red) and their accompanying DEG examples (red); down-regulated GO terms and pathways (blue) and their accompanying DEG examples (blue) are listed on the Ex_6_L23 synaptic components. Membrane-bound receptors and effector molecules may be located on the cell soma as well. (C) Synapse schematic of deep-layer EXT_4_L56 most significantly up-regulated GO terms and pathways (red) and their accompanying DEG examples (red) and down-regulated GO term and pathways (blue) and their accompanying DEG examples (blue).

HIPI1, *LRRK2*, *RAB27B*, *STX12*, and *CYTH1*), neurotransmission (including *GABRA4*, *CACNAG2*, *CACNG8*, *RIMBP2*, *SYT12*, and *CNRI*), and G protein-coupled receptor ligand binding (including *GRM2*, *GRM3*, *IQSEC2*, and *ERBB4*; Fig. 2B and table S4, A to C). The up-regulated pathways in these cells included oxidative phosphorylation and inflammatory stress responses (involving *RPS16*, *CD47*, *DNAJB4*, *HSPA5*, *HSP90AA1*, *HSPB1*, and *JUN*), signaling by neurotrophic tyrosine receptor kinase (NTRK) and transforming growth factor- β (TGF- β) receptors (such as *NTRK2*, *SLIT2*, *MAP1B*, *UNC13C*, *JUN*, and *FOS*), and growth factor response TGF- β /epidermal growth factor (EGF)/mitogen-activated protein kinase (MAPK) pathways (such as *CAMK2A*, *CAMKV*, *EPHA4*, and *GAP43*) (Fig. 2B and table S4, A to C). In contrast to EXT_6 L23 neurons, EXT_4_L56 cortical-projecting neurons manifested down-regulation of signaling by NTRKs (*NTRK3*, *AKAP5*, *DLGAP2*, and *SLIT2*) and neurodevelopmental genes related to axon guidance (*DAB1*, *ARC*, *SYNPO*, and *TCF12*; Fig. 2C and table S4, A to C). These neurons also exhibited up-regulation of genes related to mRNA splicing (*HNRNPK*, *SNRPC*, and *SLU7*), ubiquitin protein ligand (such as *USPS*, *UBE2K*, and *CUL1*), a stronger inflammatory response (*CD47*, *HLA-A*, *IFI6*, *RPL10*, *RPLP1*, *EEF1A2*, and *DNAJB14*), and intracellular signaling involving proinflammatory TGF- β signals (*TGFB2*, *SEMA6A*, and *SPPI*; Fig. 2C and table S4, A to C). These data provided evidence that these two classes of Ex intercortical projection neurons exhibited the largest degree of transcriptional dysregulation relative to other neuronal classes (Fig. 2A), implicating disruption in long-range cortical connectivity in ASD. Both populations also up-regulated genes annotated as proinflammatory. Yet these two clusters of Ex projection neurons in different cortical laminae displayed distinct patterns of down-regulation in genes composing biological pathways broadly related to neuronal signaling and neurodevelopment.

We next compared the transcriptional changes in both Ex cell types above to changes within INTs showing the most DEGs, the SST+ INTs (INT_5_SST). We observed a similar down-regulation of pathways related to synaptic vesicle regulation, neurotransmission, and synaptic organization (fig. S5A and table S4, A to C). However, INT_5_SST neurons showed a more prominent down-regulation of genes directly related to glutamatergic receptor function and signaling (involving *GRIA1/2/3*, *GRIN2A/3A*, *GRIN2B*, *HOMER1*, *SHANK3*, *NLGN1*, and *TIAMI*), structural and functional components of the action potential (including *SCN3B*, *ANK2*, *SCN1A*, *SCN8A*, and *SCN2A*), and mRNA splicing, including members of the RBFOX and STAR families (such as *RBFOX1*, *SLM1*, and *SLM2*), which play critical roles in synaptic connectivity in these cells in mouse (42, 43) (fig. S5A and

table S4, A to C). The up-regulated genes in SST+ INTs overlapped with several pathways up-regulated in EXT_4_L56 cortical projection neurons, including increases in inflammatory response (*HLA-A*, *B2M*, and *IGF1R*) and protein modification, as well as an increase in transport along cytoskeleton and calcium-mediated axon guidance related genes (*RAC1*, *RDX*, *SCNA*, and *MAPT*; fig. S5A and table S4, A to C). Mapping the SST+ INT cluster-specific down-regulated DEGs onto the spatial transcriptomic dataset demonstrated that this DEG signature was most significantly altered within superficial layers in the ASD cortex, suggesting that SST+ INTs in these laminae were the most affected (fig. S5A; right panel, $P = 0.007$ in L2/3, linear mixed model; methods).

Glial changes

Glial cells in ASD were primarily characterized by changes in the state of these cells from a homeostatic, or resting, state to reactive or activated inflammatory states, compared with controls. This change was illustrated by up-regulation of cell type-specific responses to inflammatory and interferon signaling and a parallel decrease in expression of homeostatic genes (21, 26). We describe in the following paragraphs the ASD reactive inflammatory states in the major glial types: astrocytes, microglia, and ODCs.

Genes up-regulated in protoplasmic (PP) gray matter astrocytes (ASTRO1) represent astrocyte proliferation (including *FOXO3*, *RFX7*, and *ZEB1*) and likely lie downstream of TGF- β and interferon signaling through major histocompatibility complex (MHC), other neural-immune receptors (including *HLA-B*, *HLA-C*, *HLA-E*, *CD44*, *IFI6*, *IFITM2*, *IFGR1*, *TGFB2*, and *BMP7*), and their inflammatory response effectors (including *IRF1*, *JAK3*, *MED12L*, *SMARCA2*, *STAT1/3*, and *SMAD1/3*; Fig. 3A and table S4A) (44, 45), which are up-regulated in parallel. Fibrous (FB) astrocytes (ASTRO3) exhibited similar changes, such as up-regulation of MHC and other receptors (including *HLA-B*, *HLA-C*, *HLA-E*, *CD99*, *CD44*, and *NTRK2*), but differed by up-regulation of cytoskeleton morphogenesis and focal adhesion signaling (genes including *ITGB5*, *ITGAV*, *TAB2*, *KALRN*, and *NAV3*), suggesting changes in morphology or projections from white matter FB astrocytes in the ASD cortex. FB astrocytes (ASTRO3) also up-regulated genes related to stress-related apoptosis and unfolded protein response (involving *DNAJA4*, *HSP90AA1*, *HSPA1B*, *BAG3*, and *DAB2IP*) (Fig. 3B and table S4A). Both PP and FB astrocytes exhibited a down-regulation of lipid and cholesterol biosynthesis pathway genes (*HMGCS1*). However, PP astrocytes had down-regulated growth factor and NTRK signaling genes, whereas the opposite was true for FB astrocytes in ASD relative to controls. Specifically, FB astrocytes up-regulated expression of TGF- β signaling

(*TGFB1*), growth factors (*VEGFA*), and extracellular matrix proteins (including *ADAM12* and *ADAMTS17*) (Fig. 3B and table S4A).

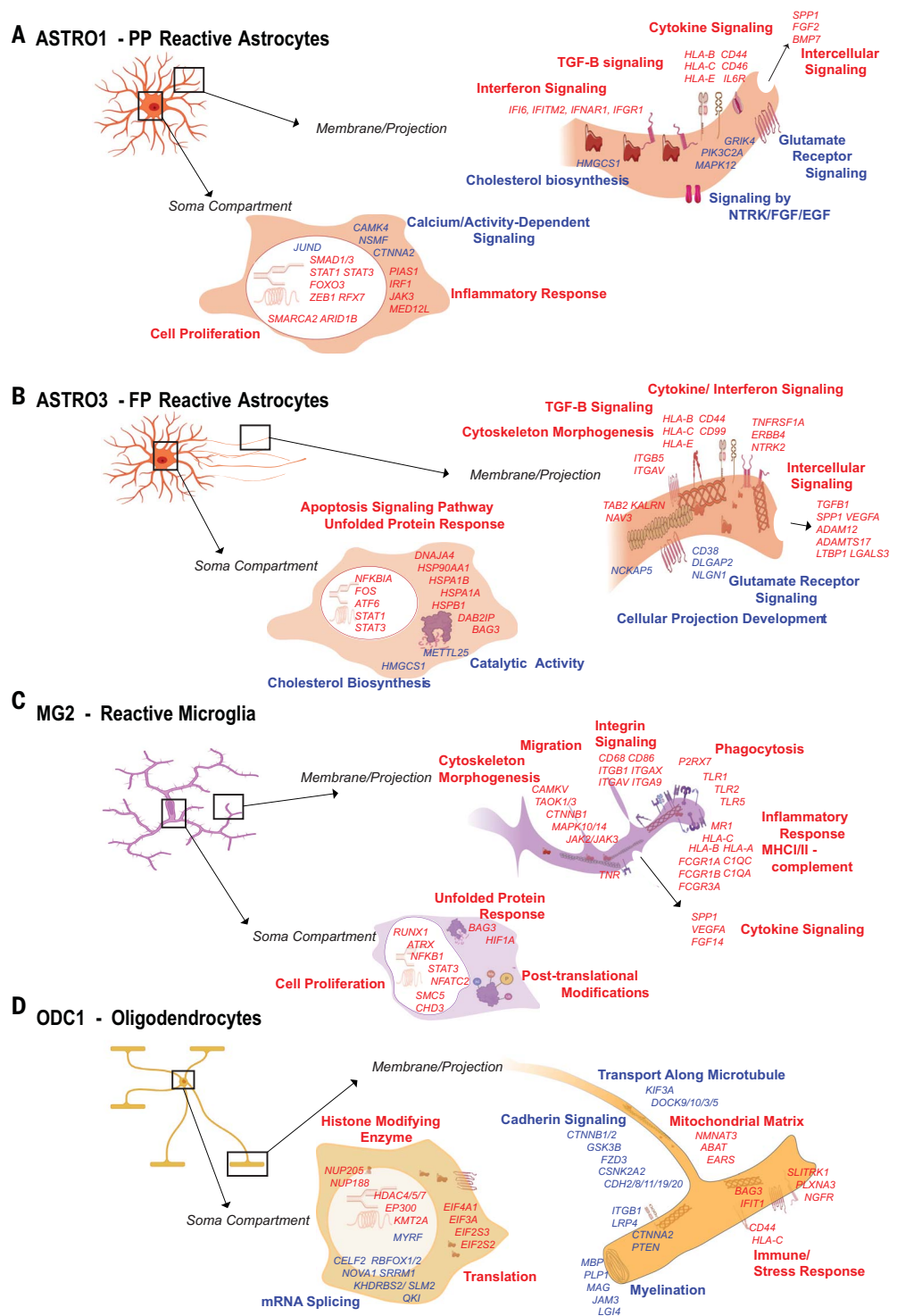
Microglia (MG2) manifested an increase in reactive-state markers, as highlighted by the up-regulation of known phagocytotic genes (such as *P2RX7*, *TLR1/2/5*, and *CD68*), including pronounced up-regulation of the extracellular signaling molecule, SPP1 (46) (Fig. 3C). MG2 cells also expressed a higher number of MHC I and II class molecules, complement, and receptors (involving *HLA-A/B/C*, *FCGR1A/1B/3A*, *CIQC*, *CIQA*, *TNR*, and *TGFB1*), more so than the other glial cell classes. In ASD, reactive MG2 cells also increased expression of genes related to cell morphogenesis (including *CAMKV*, *TAOK1/3*, *CTNBN1*, *MAPK10/14*, and *JAK2/3*) and integrin signaling (including *CD68*, *CD86*, *ITGB1*, *ITGAX*, *ITGAV*, and *ITGA9*), which suggests an increase in migration and formation of contacts with other cells (Fig. 3C and table S4A). Since many of these changes are thought to be downstream of cytokine signaling, we hypothesize from these data that SPP1 may be acting as an extracellular element responsible for coordinating the cell-cell interactions putatively directing synaptic and/or cellular removal in activated MG (46, 47).

We also observed that ODCs have shifted from a homeostatic state to a more immunoreactive state in ASD, another glial change that has not previously been recognized. For example, oligodendrocytes in cluster ODC1 up-regulated genes related to inflammatory and energy stress response (involving *IFIT1*, *BAG3*, *ABAT*, *EARS2*, *NMNAT3*, *EIF4A1*, and *EIF34*) and histone modifying enzymes (involving *HDAC4/5/7*, *KDM5A*, *EP300*, and *KMT2A*) (Fig. 3D and table S4A). However, ODC1 also contained a substantial number of down-regulated genes, the largest of any glia cell type and similar in magnitude to that of superficial L23 Ex neurons (Fig. 2B). Enriched down-regulated pathways included cadherin signaling (including *CTNBN1/2*, *GSK3B*, *FZD3*, and *CDH2/8/11/19/20*), transport along microtubule (including *KIF3A* and *DOCK9/10/3/5*), and myelination (including *MBP*, *PLP1*, *MAG*, *JAM3*, *LGI4*, and *MYRF*), which predict a potential reduction in the capability of ODC1 cells to mature and functionally extend their processes, adhere, and properly myelinate neuronal axons (Fig. 3D and table S4A).

Defining the regulatory networks underlying transcriptional changes

We next systematically integrated the cell type-specific ASD DEG data with TF networks to discover candidate drivers of the observed cell type-specific transcriptional changes in ASD relative to controls and to determine whether these could be related to causal genetic factors (Fig. 4A). We first analyzed the single-cell transcriptomes with SCENIC (48, 49),

Fig. 3. Glial cell type-specific transcriptional changes. (A) Cartoon of PP reactive astrocyte (orange), ASTRO1; black box indicates putative location of the DEGs in enlarged astrocyte membrane and projection (top) and soma compartment (bottom). We illustrate the most significant up-regulated GO terms and pathways (red) and their accompanying DEGs (red) and down-regulated GO terms and pathways (blue) and their accompanying DEG examples (blue). (B) Cartoon schematic of white-matter reactive astrocyte (orange), ASTRO3. The most significant up-regulated GO terms and pathways (red) and their accompanying DEG examples (red) and down-regulated GO term and pathways (blue) and their accompanying DEG examples (blue). (C) Cartoon of reactive MG2, the most significant up-regulated GO terms and pathways (red) and their accompanying DEG examples (red) and down-regulated GO term and pathways (blue) and their accompanying DEG gene examples (blue) are presented.



which identifies TF networks driving cell type-specific expression changes. The networks, called regulons, consist of the TF and its downstream target genes [which are significantly coexpressed by $FDR < 0.05$ and harbor the TF motif within 10 kb of the transcription start site (TSS)]. We first scored each regulon across cells to identify which regulons were preferentially associated with ASD, and then we determined

the extent of overlap of each regulon with cell type-specific DEGs (Fig. 4A; methods).

We found that 217 of 337 regulons detected were predicted to be altered in ASD in at least one cell type (table S5, A and B). We summarized the top ASD regulons predicted to regulate the most DEGs by cell type: L23 Ex neurons, SST+ INTs, and MG2 (Fig. 4, B to D; fig. S6; and table S5, C to E; methods). We found that

CUX1 regulons were a major driver of ASD DEGs: CUX1 was predicted to regulate 40% of down-regulated genes within L23 Ex neurons ($OR = 2.2$, $P = 6 \times 10^{-47}$, hypergeometric test) and 44% of down-regulated genes with SST+ INTs ($OR = 1.7$, $P = 2 \times 10^{-21}$, hypergeometric test; Fig. 4, B and C, and fig. S6). Additionally, if we considered the overlap of the top six regulons within down-regulated

A Gene Regulatory Network (GRN) Discovery

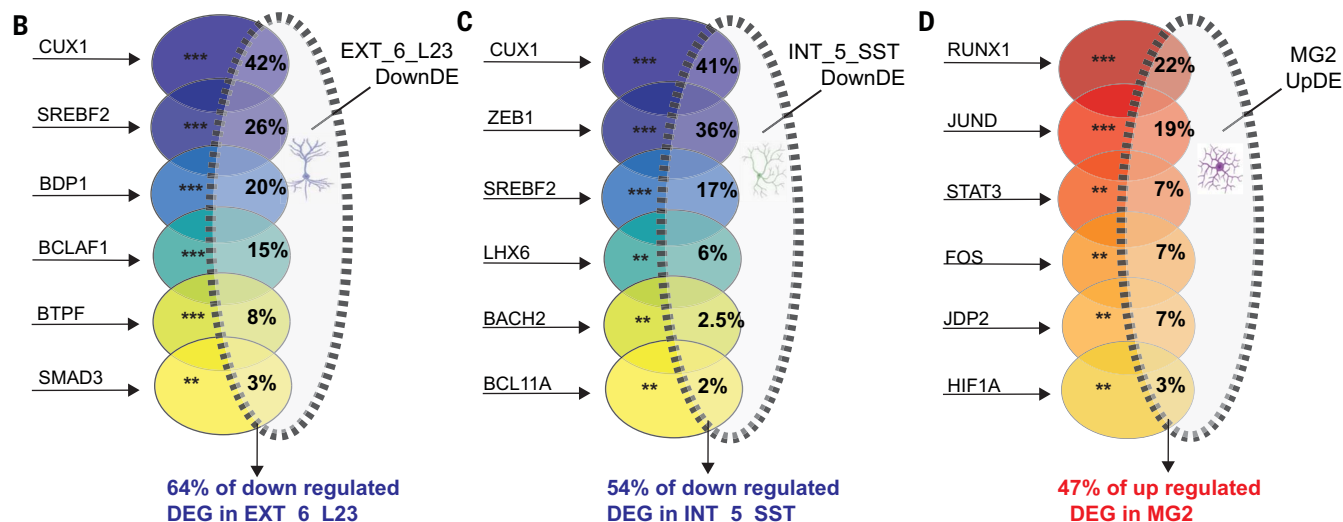
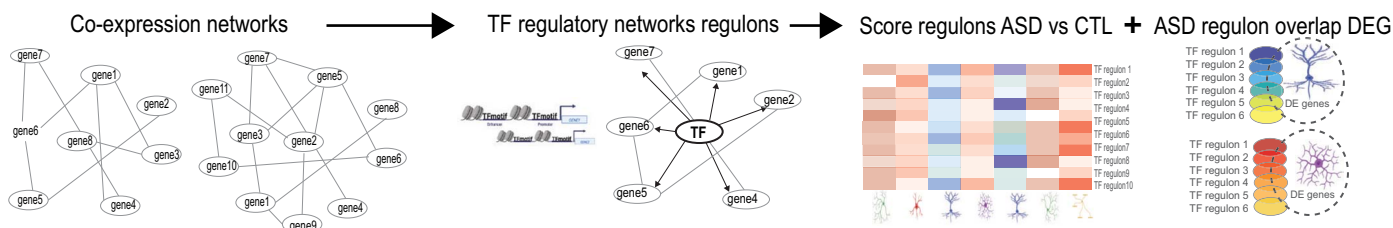


Fig. 4. Regulatory networks driving cell type-specific transcriptional changes.

(A) Schematic of network discovery using SCENIC. Initial coexpression networks were generated from cellular transcriptional profiles and then pruned to TF-driven regulatory networks, called regulons, consisting of the TF and its downstream target genes (that were significantly, FDR < 0.05, coexpressed and harbor the TF motif within 10 kb of TSS); regulons were then scored to identify which were changed within ASD, followed by assessing their overlap with cell type-specific DEGs. (B) Percent overlap between EXT_6_L23 down-regulated

DEGs and regulons predicted to regulate DEGs [highest percentage overlap (blue) to lowest percentage overlap (yellow)]. These regulons together account for 64% of down-regulated DEGs in EXT_6_L23. (C) Overlap of INT_5_SST down-regulated DEGs and largest regulons predicted to regulate DEGs (blue to yellow). Together they account for 54% of down-regulated DEGs in SST+ INTs. (D) Percentage overlap between MG2 up-regulated DEGs and largest regulons (red to yellow). Together they account for 47% of up-regulated genes in MG2 cells. *0.01 < P < 0.05; **0.001 < P < 0.01; ***P < 0.001.

genes in each neuronal cell type, we found that, collectively, they were predicted to regulate 64% of the down-regulated genes within L23 Ex neurons (involving *CUX1*, *SREBF2*, *BDP1*, *BCLAF1*, *BTPF*, and *SMAD3* regulons) and 54% of down-regulated genes within SST+ INTs (involving *CUX1*, *SREBF2*, *ZEB1*, *LHX6*, *BACH2*, and *BCL11A* regulons; Fig. 4, B and C).

The *CUX1* down-regulated regulons within both neuronal types formed direct protein-protein interaction (PPI) networks, which were also significantly enriched for ASD-associated risk genes (OR = 3, $P = 1 \times 10^{-46}$, and OR = 2.5, $P = 1 \times 10^{-20}$; hypergeometric tests), including a number of TFs that were also regulon drivers (*TAF1*, *SREBF2*, *KDM5A*, and *BCL11A*; Fig. 5, A and B; methods). The *CUX1* regulon within EXT_6_L23 was composed of genes that function within the Golgi apparatus and nucleus, related to catalytic activity on a protein (including *SYNJ2*, *MTOR*, *CYTH1*, *PSEN1*, *PLCB4*, *IQSEC2*, *ERBB4*, *SFPQ*, and *HIP1*) and gene expression (including *EP300*, *MED27*,

SMAD3, *SREBF2*, *NPAS2*, *NCOA3*, and *NCOR2*; Fig. 5, A and C). The *CUX1* regulon set of down-regulated genes within SST+ INTs was enriched for genes that function in synaptic signaling, such as calcium and potassium voltage-gated ion channels (involving *CACNA1B*, *CACNA1A*, *KCNQ3*, *CACNA1D*, *KCNC2*, and *CACNA1C*) and guanosine triphosphatase activity (involving *TIAMI*, *RASGRF2*, *FAM49B*, *KALRN*, *AUTS2*, and *STMN1*; Fig. 5, B and D). The spatial distributions of down-regulated genes from the L23 Ex neuron and SST+ INT regulons indicated that these networks were most reduced in layers 2 and 3 in ASD (Fig. 5, E and F; EXT_6_L23 regulons: *CUX1* down DEG, $P = 1.01 \times 10^{-142}$; L2/3 ASD versus CTL, $P = 8.92 \times 10^{-5}$; *CHD2*, $P = 1.40 \times 10^{-30}$; L2/3 ASD versus CTL, $P = 0.002$; *SREBF2*, $P = 1.55 \times 10^{-44}$; L3/4 ASD versus CTL, $P = 0.0003$; L2/3 ASD versus CTL, $P = 0.0004$; *FOXPI*, $P = 6.61 \times 10^{-12}$; L3/4 ASD versus CTL, $P = 0.004$; L2/3 ASD versus CTL, $P = 0.013$; INT_5_SST regulons: *CUX1*, $P = 9.11 \times 10^{-8}$; L3/4 ASD versus CTL,

$P = 0.0002$; L2/3 ASD versus CTL, $P = 0.0002$; *CHD2*, $P = 6.63 \times 10^{-66}$; L3/4 ASD versus CTL, $P = 0.001$; L2/3 ASD versus CTL, $P = 0.001$; *BCL11A*, $P = 4.76 \times 10^{-12}$; L3/4 ASD versus CTL, $P = 0.001$; L2/3 ASD versus CTL, $P = 0.006$; *CELF4*, $P = 9.16 \times 10^{-11}$; L3/4 ASD versus CTL, $P = 0.008$; L2/3 ASD versus CTL, $P = 0.021$; *FOXPI*, $P = 4.69 \times 10^{-7}$; L3/4 ASD versus CTL, $P = 0.0062$; L2/3 ASD versus CTL, $P = 0.0093$; linear mixed model; methods).

We summarize the top regulons predicted to regulate reactive MG2 DEG in ASD (Fig. 4D; fig. S6; and table S5, A and E), the top six of which account for up to 47% of up-regulated DEGs (Fig. 4D). Of note, we found that *RUNX1* and *JUND* regulons account for 21% (Fig. 4D; OR = 1.6, $P = 2 \times 10^{-10}$, hypergeometric test) and 19% (Fig. 4C; OR = 14, $P = 3 \times 10^{-7}$, hypergeometric test), respectively, of up-regulated genes within MG2 in ASD (Fig. 4D). The *RUNX1* regulon of up-regulated DEG formed a distinct, small PPI network including a few other regulon drivers (including *KDM5A* and *RUNX2*;

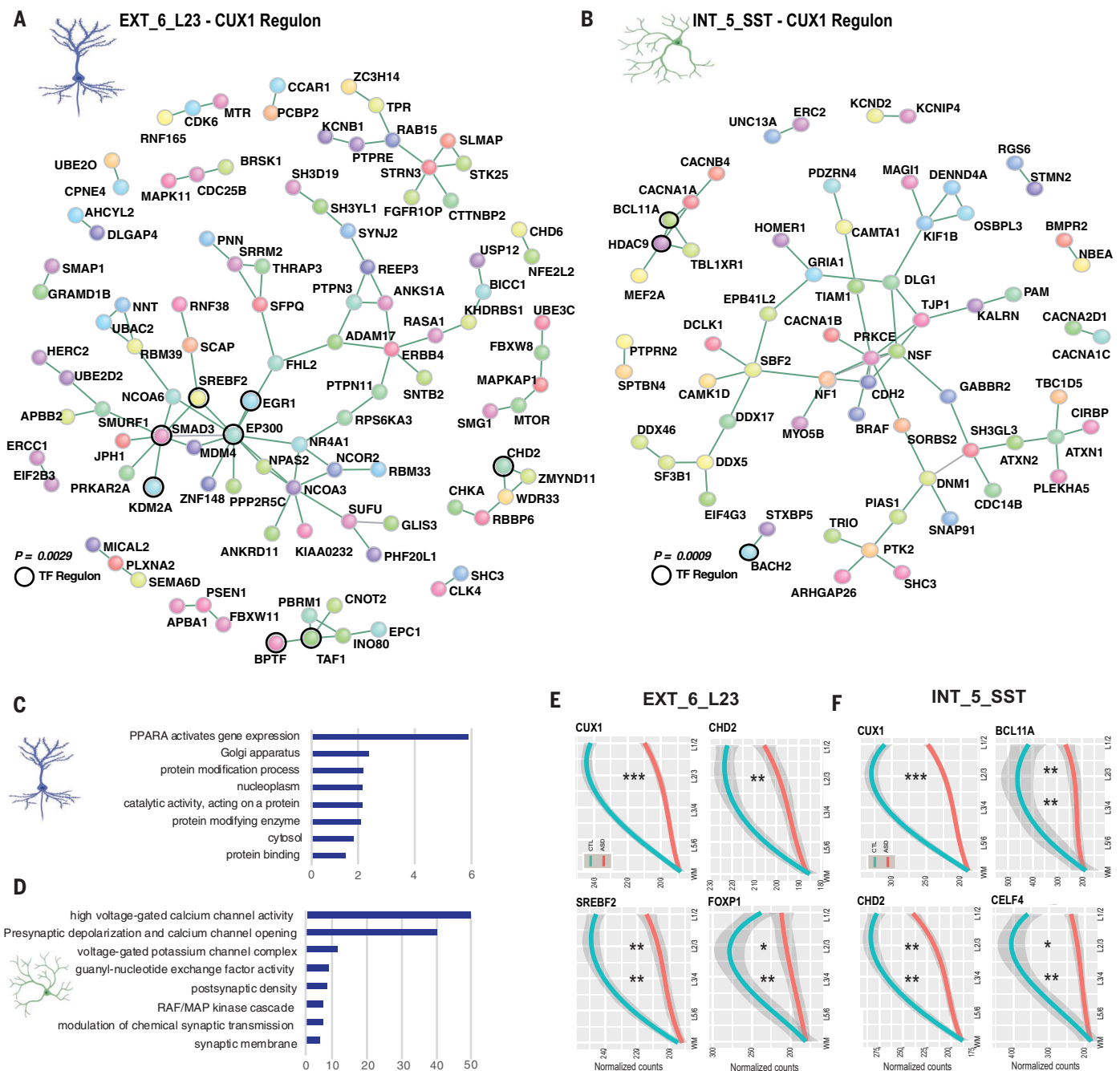


Fig. 5. CUX1 regulon in neurons forms PPI network containing ASD risk genes and regulon TF drivers. (A) PPI network formed from the CUX1 down-regulated DEGs in EXT_6_L23 cells ($P = 0.0029$, 1000 permutations) illustrating an abundance of ASD susceptibility genes and other TF drivers of regulons (black outlined circles) predicted to functionally interact. (B) PPI network formed from the CUX1 down-regulated DEGs specifically in INT_5_SST cells ($P = 0.0009$, 1000 permutations) illustrating a similar abundance of ASD genes and other TF drivers of regulons (black outlined circles) predicted to interact. (C) Significant GO terms (FDR > 0.05) associated with CUX1 regulon genes within EXT_6_L23 DEGs from all GO categories.

fig. S7A). The MG2 *RUNX1* regulon included up-regulated genes that act in the complement cascade, phagocytosis, and stress response (involving *TLR1/2/5*, *FCGR1A*, and *HIF1A*; fig. S7B) and also a sizable proportion of down-

regulated genes related to MG homeostasis (including *CX3CR1*, *CSF1R*, *LTC4S*, *PDE3B*, *SLCO2B1*, and *ADGRG1*; fig. S7C and table S5E). Notably, *RUNX1* is induced by TGF- β signaling (50), and *TGFB1* was increased in

FB reactive astrocytes (Fig. 3B), whereas *TGFB2* was increased in L56 Ex neurons (Fig. 2B). *TGFB* receptors were also increased in MG (Fig. 3C) and most other neurons (table S4A). In parallel, *SPPI* expression was increased across

(D) Significant GO terms (FDR > 0.05) associated with CUX1 regulon within INT_5_SST from all GO categories. (E) Normalized count of EXT_6_L23 down-regulated genes that overlap regulons CUX1, CHD2, SREBF2, and FOXPI, showing that regulons are located in superficial layers of CTL cortex (blue line) and are reduced most significantly in the ASD superficial layers, as expected (red line). (F) Normalized count of INT_5_SST down-regulated DEGs that overlap CUX1, CHD2, BCL11A, and CELF4 regulons are found located highly in superficial layers of CTL cortex (blue line) and reduced most significantly in the ASD superficial layers (red line). * $0.01 < P < 0.05$; ** $0.001 < P < 0.01$; *** $P < 0.001$.

these same cell types, which is predicted to promote proinflammatory, phagocytic neuron–microglia interactions (47). Moreover, TGF β 1 signaling has been shown to activate *JUN/FOS* in neurons as part of the cytokine inflammatory pathway (51). Consistent with this, we observed *JUN/FOS* expression and other pathways associated with TGF- β signaling increased in L23 neurons as well (Figs. 2, B and C, and 3, A to D, and fig. S5, A and B), implicating this pathway in these cell–cell interactions centered on MG.

To validate these network predictions, we next tested whether modulating TF expression changed expression of regulon target genes using publicly available experimental data from the Connectivity Map (CMAP) Library of Integrated Network-Based Cellular Signatures (LINCS) resource [https://clue.io/data/CMAP2020#LINCS2020 (52); methods], which is a database of standardized experimental knockdowns with short hairpin RNA (shRNA) and CRISPR knockouts of >7000 genes in multiple cell types. These analyses, which permitted testing of >130 TFs to target gene relationships of regulons, showed significant agreement between our predicted regulatory networks and experimental RNA-seq knockout or knockdown of regulon TF drivers in neural cells, providing external experimental validation of the regulon networks (P range = 1.76×10^{-4} to 4.86×10^{-13} permutation, see methods; fig. S8, A and B). We also leveraged a published Perturb-seq dataset in K562 cells to test 10 overlapping TFs from both datasets [(52); methods], which also confirmed that their knockouts resulted in significant changes in the predicted regulon target genes ($P = 1.45 \times 10^{-6}$ permutation, see methods; fig. S8C). These findings support our regulon network relationships between TFs and downstream genes across a range of cell types and experimental manipulations.

snATAC-seq and TF footprinting validate regulatory network predictions

Although SCENIC predictions have been shown to be reliable in predicting experimentally validated regulatory networks (1, 53–55) and, moreover, were supported by the in vitro experimental perturbation data above, we next sought to further validate regulatory changes of TFs. We investigated chromatin accessibility by performing snATAC-seq from a subset of the same 22 samples (12 cases and 10 controls) subjected to scRNA-seq (methods; table S6A). We generated >100,000 single-nucleus profiles after quality control filtering, which we integrated with the snRNA-seq to verify and validate cell type cluster annotations (Fig. 6A and fig. S9, A and B; methods). We observed a high concordance between the snATAC-seq and snRNA-seq clusters, as evidenced by high correlations between cluster expression profiles (table S6B; methods). Moreover, we assessed the overlap

between snATAC-seq differential chromatin accessibility and snRNA-seq-derived gene expression changes when there were sufficient numbers of cells and gene expression changes for adequate power, finding significant overlap for most cell types as expected (figs. S10, A to E, and S11, A to E; P range = 1.1×10^{-210} to 0.04, hypergeometric test).

We next leveraged the open chromatin profiles from each major cell type to assess changes in TF binding on the basis of DNA footprinting using TOBIAS (56, 57) (Fig. 6B; methods). DNA footprints overlapping with TF motifs were defined as direct TF binding sites. TF binding activities were then quantified by scoring each footprint's depleted signal at its predicted binding site relative to the surrounding background chromatin accessibility (methods). Using this approach allowed us to more accurately recover the binding activities of hundreds of TFs at a time in comparison to other technologies that permit assessment of the activity of one TF at a time (56). We focused on TFs predicted to account for the major changes in DEGs described above (Figs. 2 to 4 and fig. S6). Within Ex L23 neurons, we observed that 48% of the transcriptionally down-regulated TFs predicted to drive downstream DEGs within the network in ASD have changes in binding occupancy within the relevant TF DNA footprints (table S7A). For example, in EXT_6_L23 neurons, we saw a decrease in the footprint activity of *BCL6* ($P = 2.45 \times 10^{-114}$), *TCF3* ($P = 1.28 \times 10^{-63}$), and *MAFB* ($P = 9.12 \times 10^{-100}$) as well as an increase in binding for up-regulated TFs such as *FOSB* ($P = 1.40 \times 10^{-195}$), *JDP2* ($P = 1.24 \times 10^{-196}$), and *SMAD5* ($P = 4.16 \times 10^{-128}$; Fig. 6C) with examples of altered footprints for *MEIS3* and *ELK1* (Fig. 6D, footprints, and table S7A). Likewise, within reactive MG2, we validated the changes in target binding for 43% of transcriptionally altered TFs by footprinting (table S7B). We observed increases in TF activity for *IRF7* ($P = 2.35 \times 10^{-132}$) and *POU2F2* ($P = 5.37 \times 10^{-107}$) (Fig. 6E and table S7B) with examples of altered footprints for *JUND* and *NFKB1* (Fig. 6F) in MG2. Altogether, we found that the SCENIC regulatory networks and DEGs were mostly supported by TF occupancy changes within open chromatin in ASD relative to controls, which, when combined with the perturbation experiments (fig. S8), provide strong orthogonal experimental support for regulon activity changes in ASD.

ASD risk gene enrichment

To determine whether these regulons represent known causal aspects of ASD biology, we next tested whether the drivers or their downstream gene targets were enriched in known ASD risk genes (SFARI syndromic, tiers 1 and 2; methods). Indeed, 30 of the regulons dysregulated in ASD brain were driven by high-confidence TF risk genes from the SFARI

database, a significant enrichment (including *CUX1*, *CHD2*, *FOXP1*, *BCL11A*, and *SOX5*; fig. S12, rare, OR = 1.64, $P = 0.004$, hypergeometric test). Further, 21 of these regulons were enriched for genes harboring common genetic susceptibility for ASD (37) (including *CELF4*, *SREBF2*, *ETV5*, and *NR3C1*; $P = 0.049$ to 0.0007; methods; fig. S12, common; table S7C). These regulons are primarily active within Ex and INT neuron populations (highest in Ex neurons) and less so in glia, consistent with previous studies of ASD susceptibility gene expression patterns (36, 53) (Fig. 6G and fig. S12). However, we note that there were several TF-risk regulons (e.g., *SOX5*, *SOX6*, *TCF7L2*, *PAX6*, and *SREBF2*) differentially expressed in ASD ASTRO1, homeostatic MG1, and OPCs, in addition to neurons (Fig. 6G and fig. S12). ASD risk gene enrichment has not been demonstrated before in glial cells. These regulons comprised genes that were down-regulated, which was the expected direction given the observed down-regulation of TF drivers, and which would occur subsequent to haploinsufficiency of a risk gene, which is predicted for most of these genes in ASD.

We also examined whether these regulons' target genes were also enriched for ASD susceptibility genes (SFARI tier 1, 2, or syndromic) (fig. S13A), as well as genes associated with intellectual disability (ID) (58) and fragile X protein (FMRP) targets (59). Indeed, many regulons driven by ASD TF susceptibility genes, or those enriched for ASD-common variation, were also enriched for ASD risk genes ($n = 19$) and FRMP targets ($n = 17$), whereas only a few were enriched for ID risk genes ($n = 5$; Fig. 6H) relative to other regulons. Moreover, networks similarly generated from published single-cell transcriptome data in schizophrenia (60) lacked enrichment for ASD genes (fig. S13B). Taken together, these data provide a critical starting point for understanding the alterations in cellular and circuit mechanisms driven by specific risk genes in the brains of individuals with ASD.

Discussion

In this study, we performed a comprehensive single-cell genomic assessment of the frontal cortex in individuals with ASD, substantially extending previous observations and providing important insights into ASD biology. The depth and breadth of this single-cell analysis, involving large numbers of individuals and cells profiled, solidifies the picture of the major cortical cell types affected in ASD, including alterations in cell type composition, remodeling of cell states, and the identification of cell type-specific transcriptional cascades. We also provide a reliable, comprehensive list of DEGs, which showed high correspondence to orthogonal bulk and spatial transcriptomic data and demonstrated parallel changes at the cellular level in a major genetic

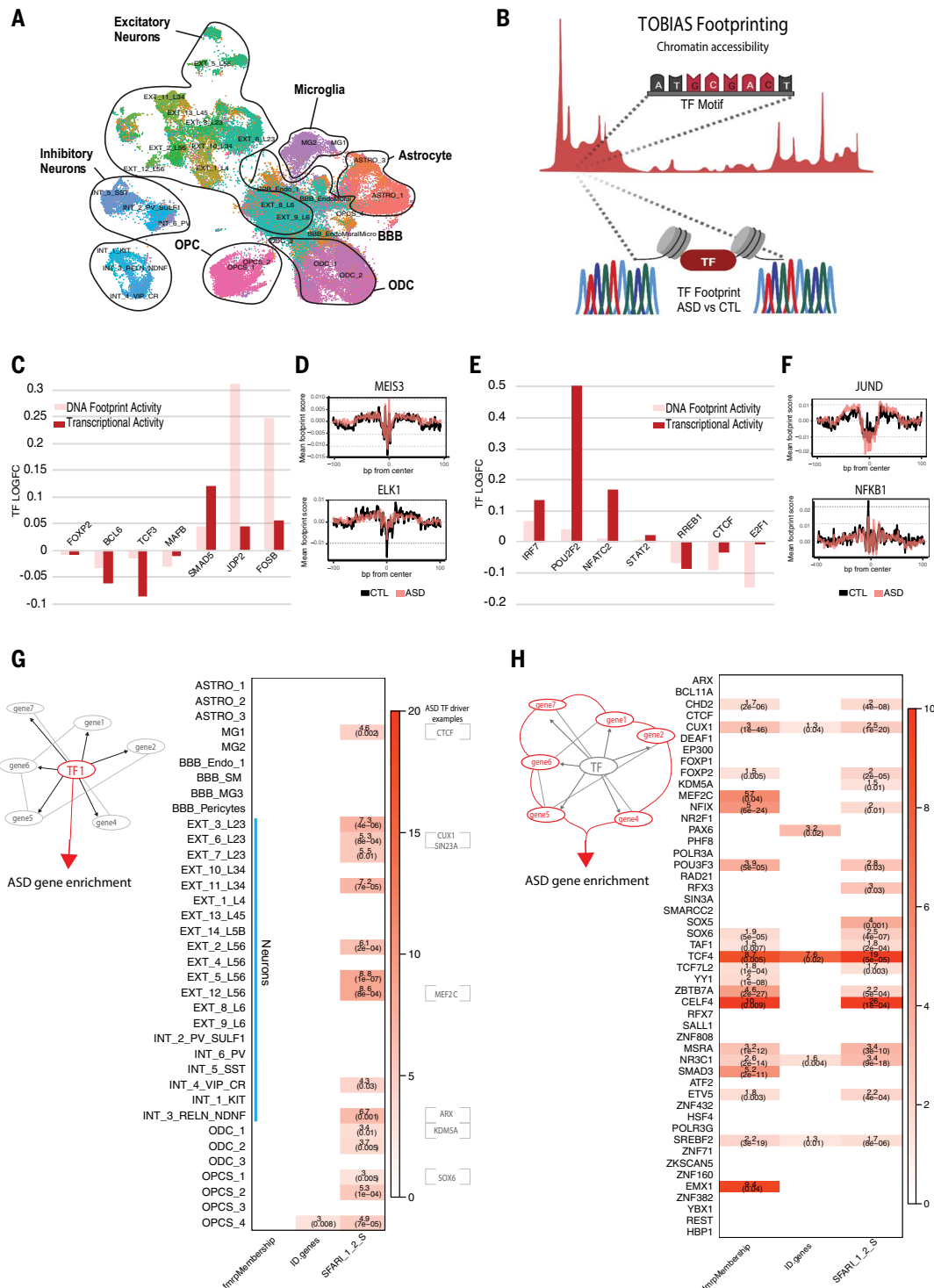


Fig. 6. Neuronal cell classes and homeostatic glia types are enriched for regulons driven by ASD risk TF genes. (A) UMAP of cellular clusters from scATAC-seq computationally annotated by scRNA-seq clusters. (B) Footprinting approach with TOBIAS to assess changes in TF occupancy within open chromatin profiles of each cell type. (C) Bar plot of representative TF occupancy and TF expression logFC of ASD versus control in open chromatin (light red) and DEGs (dark red) of aggregated L23 Ex neurons. (D) Representative TF footprints of significantly (FDR > 0.05) down-regulated TF occupancy in open chromatin for MEIS3 (top) and ELK1 (bottom) from CTL (black trace) and ASD (red trace).

(E) Bar plot of representative TF occupancy and TF expression logFC of ASD versus control in open chromatin (light red) and DEGs (dark red) of aggregated MG2. (F) Representative TF footprint of significantly up-regulated TF occupancy in open chromatin of MG2 cells from CTL (blue trace) and ASD (red trace) brains for JUND (top), and NFKB1 (bottom). (G) ASD vulnerability gene enrichment heatmap of TF drivers by cell type. OR (red) and P value (black) included for each cell type. Neuronal clusters marked by a blue line. Example TF drivers in gray brackets. (H) Risk gene enrichment heatmap of regulon genes to gene sets associated with FMRP, ID, and ASD risk. OR (red) and P value for each regulon.

form of ASD, (dup)15q11-13 syndrome. The integration of snRNA-seq, snATAC-seq, and layer-specific spatial transcriptomics allowed us to describe and validate candidate regulatory networks driving cell type-specific transcriptional changes and their location within cortical laminae. Although we observed broad transcriptional changes in both inhibitory and excitatory neurons, consistent with previous observations in bulk tissue, the magnitude of changes varied considerably by cell type. We found that most changes in neuronal gene expression occur in glutamatergic neurons within the superficial layers of the cortex, confirming a previous, smaller analysis implicating superficial-layer projection neurons (13). We also identified large changes in SST interneurons, and intracortical projecting L56 neurons. Moreover, these changes in neuronal gene expression in ASD were accompanied by marked shifts to more activated proinflammatory states in multiple glial cell types, including ODCs, MG, astrocytes, and cells that make up the BBB. These data are available to explore as a resource on the PsychENCODE website (67) and our ASD single-cell browser, the ASD Single-Cell Gene Expression (scGENE) Portal (<http://solo.bmap.ucla.edu/asdscgene/>).

Transcriptomic changes associated with a neuropsychiatric condition, although representing a quantitative genome-wide molecular measurement, may represent the consequence of that condition and do not necessarily imply causality. We reasoned that by identifying potential transcriptional regulators and subsequently testing whether these TFs or their downstream targets were enriched in genetic risk, we could implicate potential causal drivers. Our analysis identified candidate TF drivers that account for a substantial portion of major gene expression changes in multiple neuronal and glial cell types, of which many were validated by the CMAP LINCS resource and TF footprinting. These TFs, coupled with their downstream targets, called regulons, are enriched in known high-confidence ASD risk genes harboring rare variants and common polygenic risk score for ASD, which supports their potential causal role in ASD pathogenesis. Moreover, these analyses delineate the potential downstream consequences of these causal genetic variants in the ASD brain by identifying DEGs and the biological pathways that are regulated by these risk genes in the human brain. These regulon networks associated with ASD risk reflect down-regulation of synaptic vesicle regulation, neurotransmission, and components related to synaptic development and organization pathways, which have been observed in other studies (8, 13, 14). A number of these TFs have important roles in the development and maintenance of cell identity (e.g., *CUX1*, *SOX6*, and *FOXP2*), providing a link between fetal development, invol-

ing ASD risk-locus enrichment in specific TFs, and its postnatal ramifications, represented by cortical transcriptional changes in ASD. We emphasize that more definitive determination of the causal impact of risk genes on these transcriptional regulatory circuits awaits direct experimental perturbations, for example, in human in vitro organoid models (62–64).

These data support previous predictions of ASD risk gene enrichment in superficial-layer neurons from developing neurotypical human cortex (3, 65) and extend these observations by (i) implicating superficial SST+ INTs and deep-layer, interhemispherically projecting Ex neurons; (ii) linking regulatory networks driven by ASD risk TFs to cell type-specific DEGs in postnatal brain in ASD and applying spatial transcriptomics to corroborate the laminar distribution of these changes; (iii) suggesting that the shift in expression of homeostatic genes to genes that underlie “reactive” or “activated inflammatory” state in microglia and astrocytes is enriched in superficial gray matter layers and also extends to ODCs and cells that make up the BBB; and (iv) showing that some of these changes in microglia and ODCs are linked to multiple candidate TF drivers that are known ASD risk genes, providing a potentially causal link to some of the reactive changes in these glial cells. Further, these data support a mechanistic model for the potential downstream impact of ASD risk genes through their activity as transcriptional drivers. These TFs converge primarily in intracortical Ex projection neurons. We posit that primary Ex neuron dysfunction in conjunction with local circuit interneurons may subsequently influence microglia and astrocyte reactive states, because the latter are also more concentrated in superficial cortical laminae, where both risk genes and transcriptional changes are most prominent. These data are consistent with the framing of ASD as a developmental synaptic disconnection syndrome involving intracortical connectivity (66) but also indicate consequential local, presumably maladaptive, changes in both interneurons and glia that could be leveraged as potential therapeutic modulators.

Individuals with autism may manifest a diverse array of coexisting conditions (including epilepsy, attention deficient hyperactivity disorder, schizophrenia, gastrointestinal conditions, cerebral palsy, bipolar disorder, anxiety), traits (including apraxia or speech difficulties, social communication differences, sensory sensitivities, learning differences, specialized interests), and related behaviors (such as repetitive movements or “stimulatory-seeking,” echolalia, and more) that make up their individual characteristics, or “neurotype,” of ASD (34, 36–40, 67–69). Thus, it will require large numbers of individual samples to elucidate subgroups and to definitively match specific

patient traits, such as cognition or social behavior, to the observed molecular and cellular changes observed here. We note that our analysis included tissue from five individuals with a monogenic form of ASD, (dup)15q11-13, who, despite manifesting distinctive gene expression patterns, do share the major transcriptional alterations and pathways observed in idiopathic ASD at the bulk level (7), which we demonstrate here at the single-cell level. Notably, genes within the 15q duplication region show considerable cell-to-cell variability in their expression changes, further demonstrating the importance of understanding cell type-specific effects of genetic variation, especially in the context of major changes in gene dosage.

Finally, recent work has demonstrated cortex-wide transcriptional changes in the ASD brain (14), including primary sensory regions, which is consistent with the preferential involvement of neurons responsible for intracortical and interhemispheric connectivity that we demonstrate here. These data provide a starting point for developing mechanistic, molecular, and cellular explanations for observations of changes in both short- and long-range cortical connectivity and cortical circuits in ASD (3, 65, 70, 71). A major challenge is now to understand how these cell-cell interactions develop and evolve at both a cellular and circuit level and to determine how they relate to physiological and behavioral states.

Materials and methods

Sample acquisition and preparation for snRNA-seq and snATAC-seq

Postmortem cortical brain samples were acquired from the Autism Brain Net project [ABN, formerly the Autism Tissue Project at Harvard (ATP)] and the University of Maryland Brain Banks (UMDB) through the NIH NeuroBioBank. A total of 63 samples from subjects with ASD, dup15q syndrome, and nonpsychiatric controls from the prefrontal cortex (PFC) or from Brodmann area 9 (BA9) and to a smaller extent BA4/6. Thirty-one subjects had not been previously profiled. Three samples were dropped in the final integration. BA4/6 samples were previously subjected to bulk studies: Parikshak *et al.*, *Nature* 2016 (7); Gandal *et al.*, *Science* 2018 (8), and a small subset of samples (six samples) were processed for a different single-cell analysis in Gandal *et al.*, *Nature* 2022 (14).

Frozen brain samples were placed on dry ice in a dehydrated dissection chamber to reduce degradation effects from sample thawing and/or humidity. Approximately 20 to 50 mg of cortex was sectioned, ensuring specific gray matter-to-white matter boundary and ratio. The tissue section was homogenized in RNase-free conditions with a light detergent (0.01% TritonX100/phosphate-buffered saline (PBS)/1% bovine serum albumin (BSA)/RNase/0.01% digitonin)

briefly on ice using a dounce homogenizer, filtered through a 40- μ m filter and centrifuged at 1000g for 8 min at 4°C. The pelleted nuclei were then filtered through a two-part micro gradient (30%/50% iodixanol sucrose medium) for 20 min at 4°C. Clean nuclei were pelleted away from debris. The nuclei were washed two more times with PBS/1% BSA/RNase and centrifuged at 500g for 5 min. Cells were inspected for quality (shape, color, membrane integrity) and counted on a Countess II machine (#AMQAF1000, Thermo Fisher). They were then loaded onto the 10x Genomics platform (Chromium controller #1000204, 10x Genomics) to isolate single nuclei and generate libraries for RNA sequencing on the NovaS4 or NovaS2 Illumina machines for a depth of about 65,000 reads per cell.

snRNA-seq data processing

After sequencing, Cell Ranger software from 10x Genomics (<https://support.10xgenomics.com/single-cell-gene-expression/software>) was used to prepare fastq files, and reads were aligned to the human GRCh38 pre-mRNA genome to generate gene-by-cell matrices for each library. Pegasus (<https://github.com/lilab-bcb/pegasus>), a Python-based program, was used to stringently filter cells, remove doublets by scrublet, and integrate and batch-correct all libraries together. Initially, we performed low-resolution analysis with top 10 principal components (PCs) and removed an identified low-quality cell cluster with no canonical gene signature but high mito_percent (mitochondrial gene percentage) and low ngenes (number of genes per cell). Then we reran the analysis with top 65 PCs after removing these cells. Cells were filtered or removed if they expressed >6000 or <250 genes or if they had >10% mitochondrially mapped reads. Dimensional reduction was done by principal components analysis (PCA) with n_genes and percent_mito regressed out. Harmony (as part of the Pegasus suite) was used to integrate and batch-correct libraries. Louvain clustering was performed to cluster the cells and visualize resulting clusters with uniform manifold approximation and projection (UMAP).

Cell type annotation

Cell types were annotated on the basis of expression of known marker genes visualized by UMAP and by performing unbiased gene marker analysis (table S1C). Canonical genes were selected on the basis of mouse and human studies as well as published reference atlas enriched genes (15–18). We integrated the Velmeshev *et al.* (13) dataset with our clusters through Seurat's R-based program (72) (<https://satijalab.org/seurat/index.html>), implementation of FindTransferAnchors and TransferData functions as detailed in vignettes online (https://satijalab.org/seurat/articles/integration_mapping.html). In brief,

the functions aim to build cell pairwise correspondences between single cells across the reference datasets and query datasets using PCA coordinates, thus allowing integration of datasets into a shared space and serving as an effective matching tool of one cluster to another on a cell-by-cell basis. We also used Azimuth, a web-based portal of the same Seurat functions above, from the Allen Brain Atlas (19) (<https://azimuth.hubmapconsortium.org>) to align our cell clusters to the human and mouse motor cortex clusters [BICCN (18)] (table S1D).

Cortical cell composition analysis

A cell type-by-subject count matrix of cell number per subject per cluster was generated from metadata. Data were then transformed to cell proportions by total cells for each library and log normalized. We then used the R-package LimmaVoom (73) to stringently ask whether any clusters were significantly over- or under-represented ($FDR \leq 0.1$) in the ASD subjects in comparison to CTL subjects by implementing a linear model (co-variables were included if a Spearman correlation was >0.25 against the top five clustering PCs): ~diagnosis + Chemistry10X + age + PMI (post mortem interval) + totalReads (table S3).

Layer-specific spatial transcriptomics dataset (NanoString Technologies)

We selected four CTL and four ASD subject samples to embed frozen cortex overnight in OCT [Tissue-Tek Cryo-OCT Optimum Cutting Temperature (OCT) #M-649078-1412] at –80°C. We then equilibrated blocks to –20°C in a Leica cryostat (# CM1520) and collected four 10- μ m-thick sections (at least 150 μ m apart from one another within the cortex) onto NanoString slides. NanoString Technologies stained slides with DAPI (4',6-diamidino-2-phenylindole) and GFAP (glial fibrillary acidic protein) to help mark white matter boundaries from gray matter in layer 6 and layers 1/2. The sections were then treated with ultraviolet (UV) light–photocleavable oligo-probes of about 18,000 genes from the human genome. We assisted the NanoString technician, J. Gong, with choosing regions of interest (ROIs), and we created three or four ROIs within gray matter for L1/2, L2/3, L3/4, L5/6, and three ROIs within white matter. The ROIs were then exposed to UV light, which cleaves the probes, releasing them for subsequent RNA sequencing. After sequencing, the processing and normalization of data involved matching the raw probe counts to sequencing results to filter out unused probes and remove outliers, and then the data were normalized to the third quartile.

Mapping gene activity signatures to layers

To look for spatial activity across the layer dataset using cell type annotation markers and/or

ASD DEGs, we first filtered each cell type-specific gene set or signature into a small representative list containing only positive genes specific to that set of comparisons. For cell type-specific ASD DEGs, only significantly down-regulated genes ($FDR < 0.05$ and $\log FC > 0.2$) were used to query layer activity for interneurons. For cell type-specific regulons, the top DEGs per regulon per neuronal type and the top up-regulated regulon specific to MG2 were used to query layer count distribution. Linear mixed models were used to assess diagnosis effects and diagnosis and layer interaction effects using individual and nGene as random effect covariables. Post hoc tests were assessed using estimated marginal means (least-squares means).

Pseudobulk expression analysis by cell type

We generated pseudobulk counts for each sample by adding counts from the same cell type. Then pseudobulk counts were normalized by a variance-stabilizing transformation method. To identify genes differentially expressed in ASD compared with control in each cell type, we examined covariates with top five PCs from normalized pseudobulk expression matrix. We identified the following covariates consistently correlated with top five PCs for each cell type: age, PMI, BrainRegion, SeqBatch, Mito_perc, and ngenes. We then randomly selected subjects 500 times and calculated average beta to regress out effects of these covariates. Then we used limma-voom to identify DEGs for each cluster.

Pseudobulk RNA-seq “bulk” library

To correlate bulk tissue mRNA levels with nuclear RNA levels, pseudobulk data were aggregated all nuclear profiles by sample, and we compared bulk mRNA (7) FPKMs (fragments per kilobase million mapped) to normalized UMIs. We examined co-variables with top 5 PCs from the normalized pseudobulk matrix. Then we used limma-voom to identify DEGs in ASD versus CTL. We identified the following covariates consistently correlated with top five PCs for each cell type (age, PMI, SeqBatch, Mito_perc, and ngenes) and bootstrapped 500 subjects randomly to calculate average beta to regress out effects of these covariates.

Statistical tests

Unless otherwise stated, statistical tests were run using R (4.1.0). Only *P* values with post-hoc FDR or Bonferroni-corrected value of 0.05 were considered significant, unless otherwise indicated.

Module enrichment

We implemented a Fischer's exact test to analyze the overlap of up-regulated and down-regulated genes for each cluster's DEG set in

ASD. We filtered the DEG lists to be $FDR < 0.05$ and the (differential expression value of genes assigned to specific modules) KME of the module genes >0.5 to best capture the single-nucleus expression signature to module relationship. Odds ratio and estimated significance of overlap were determined using a background list of genes expressed in postmortem brain.

Gene Ontology (GO)

We used PANTHER (Protein Analysis Through Evolutionary Relationships; <https://pantherdb.org/>) to perform statistical overrepresentation tests for DEGs from each cluster and for cell type-specific regulons using Fisher's exact test with Bonferroni correction. We used brain-expressed genes as background. We only report GO terms that are significant (corrected $P < 0.05$). All GO categories (cellular composition, biological process, molecular function, REACTOME pathways, protein function) are reported for each cell, and all were used to interpret DEG changes.

We also implement SynGO (Synaptic Gene Ontologies; <https://www.syngoportal.org>) under the same settings as above to verify synaptic enrichments in neurons as well as validate pre- and postsynaptic gene functions.

Protein–protein interaction networks

DAPPLE [Disease Association Protein-Protein Link Evaluator (74)] results for evaluating the significance of PPI networks of the cell type-specific regulons were all done using 1000 permutations (within DAPPLE parameter), and P values < 0.05 were considered significant.

Susceptibility gene overlap in bulk DEGs

To estimate overlap of ASD genetic risk factors with DEGs, we used the hypergeometric test with the list of DEGs in the pseudobulk library and list of all genes expressed. These lists were overlapped with all genes in the SFARI Gene Module database having evidence of type 1, type 2, or syndromic genetic association with ASD (<https://gene.sfari.org/database/gene-scoring>).

Susceptibility gene overlap in regulons

We used Fisher's exact tests to estimate the significance of overlap between genes within select regulons and significantly DEG ($FDR < 0.05$) genes by cell type, split by up-regulated ($\log_2FC > 0.2$) or down-regulated ($\log_2FC < -0.2$) using a background list of genes expressed in postmortem brain (expressed in at least 100 cells, 15,370 genes). Estimated P values were FDR corrected. Similarly, we used Fisher's exact tests to compare the genes associated with intellectual disability (58), FMRP (59), or SFARI ASD risk genes considered either level 1, level 2, or syndromic with TFs or drivers of ASD regulons within each cell type. Odds ratio and estimated significance of overlap was de-

termined using a background list of genes expressed in postmortem brain.

Common variation enrichment in regulons

Partitioned heritability of regulons and ASD common variants (37) was assessed using LD Score Regression v1.0.0 (75). Heritability was calculated by comparing the association statistics for common genetic variants falling within regulatory elements associated with specific regulon genes, with the LD-score, a measure of the extent of the LD block. First, an annotation file was created, which marked all single-nucleotide polymorphisms (SNPs) that fell within the regulatory elements for each regulon gene (10 kb up- and downstream of gene). LD-scores were calculated for these SNPs within 1-cM windows using the 1000 Genomes European (EUR) data. The enrichment coefficient was calculated as the heritability explained for ASD common variation within a given annotation divided by the proportion of SNPs in the genome and subjected to Bonferroni correction.

Loss-of-function study validations of regulatory networks and transcriptional regulation

To examine the effect of predicted regulon target genes of TF regulon drivers, we downloaded publicly accessible data from the CMAP LINCS 2020 resources (<https://clue.io/data/CMAP2020#LINCS2020>), using both level 5 shRNA knock-down (sh) and CRISPR (xpr) experimental datasets (files: level5_beta_trt_sh_n238351x12328.gctx and level5_beta_trt_xpr_n142901x12328.gctx). For central nervous system filtered comparisons, data were filtered for only “NPC” and “NEU” cell types. Using a background of post-mortem expressed genes, we randomly sampled genes >1000 permutations that were size-matched to each regulon gene set size and computed the average of the absolute z-score value for each randomly sampled gene across all manipulation experiments for each TF regulon driver (sh or xpr, separately). We then compared the distribution of the mean z-score for each randomly selected size-matched gene set across all regulons present in the dataset to the mean z-score for the regulon target predicted z-score across all regulons present in each of the datasets. P values were calculated by z-transformation of the mean of all regulon target genes compared with the distribution of randomly selected genes. We also tested whether filtering for transcriptional activity scores (TAS) of >0.2 , as calculated in the CMAP dataset (tas column), changed the effect. The same analysis was conducted for TF regulon drivers manipulated through a CRISPR knock-out strategy in K562 cells (52) using cell-state corrected values [table S4 in (52)].

SCENIC regulatory networks

We applied pySCENIC [(50); <https://pyscenic.readthedocs.io>] to a subset of cells within our

dataset (10,000 cells, randomly subsetted). We then constructed GRNs informed by a list of all transcription factors (RCIStarget). GRNs are then pruned to only include downstream targets that have a TF-binding motif within 10 kb of the TSS. Finally, area under the curve (AUC) scores were calculated across 100,000 cells, and z-scored AUC scores were calculated. Regulons were considered expressed in a cell type if they had an average z-score of >0 . AUC z-score values of expressed regulons within each cell type were compared between ASD and CTL using t tests and FDR-corrected P values (corrected within cell type on the basis of the number of expressed regulons).

Regulon overlap of transcriptional changes in each cell type

The list of TF drivers of regulons that were significantly differentially expressed ($FDR > 0.05$) in at least one cell type in ASD (217 TFs) was overlapped with each cell types' ASD DEGs to identify which TF's expression was down- or up-regulated. To determine enrichment of up-regulated and down-regulated genes within regulons, we used the fgsea (fast gene set enrichment analysis) R package (<https://bioconductor.org/packages/release/bioc/html/fgsea.html>) to determine the gene set enrichment within regulons. DEGs were ranked on the basis of logFC within cell types. Significantly enriched regulons were chosen on the basis of a filtered $FDR < 0.05$. Then we overlapped the top six largest regulons enriched in each cell type DEG to find the percentage to which they explain DEG changes and hypergeometric tests (above) to determine the significance.

scATAC-seq and data analysis

Nuclei were extracted, as above, simultaneously from 22 samples subjected to scRNA-seq—consisting of 12 ASD samples and 10 CTL individuals that were balanced by age, PMI, and cause of death. Libraries were aligned to reference genome with 10x Genomics software, Cell Ranger (cellranger-atac count; <https://support.10xgenomics.com/single-cell-atac/software>), then CTL libraries and ASD libraries were aligned together using the cellranger-atac aggr function. We used Signac [(76); <https://stuartlab.org/signac/>], an R-based program to stringently filter out low-quality cells through nucleosome banding score < 4 and TSS < 2 , leaving 41,683 ASD and 31,754 CTL high-quality cells from each dataset. We merged ASD and CTL objects together by union of peaks from both and then re-called peaks so the same set of features are accounted across both datasets. We then normalized with term frequency-inverse document frequency (TF-IDF), reduced dimensionality using partial singular value decomposition, and clustering regressing LSI1 and peak_region fragments out to generate 30 clusters. To annotate the clusters, we first created a gene activity

score matrix where gene expression levels are roughly computed from fragment counts within gene body elements (see <https://stuartlab.org/signac/articles/>). We then implemented Seurat's FindTransferAnchors function as described above. We find 30 clusters in the scATAC-seq dataset that fit the profile of the snRNA-seq clusters; given that the scATAC-seq data are ~6× smaller than snRNA-seq, it is expected to not have as many clusters, specifically they do not have cluster OPCS_3, ASTRO_2, and a number of small L56 excitatory clusters. These clusters likely did not reach enough numbers in the ATAC dataset to form a separate cluster. We then used Spearman correlation to assess how closely related the cluster-specific marker gene profiles from the snATAC-seq clusters with our snRNA-seq cluster marker gene profiles.

Peak calling and prediction of TF activity in snATAC-seq data

We merged reads from individual cells of the same snATAC clusters per sample (<https://github.com/timoast/sinto>) with the following commands: `sinto filterbarcodes-bam {bam}-cells {barcode_path}-outdir {subset_dir}-nproc 4`. Pseudobulk replicates were further created with samtools merge by disease condition. Peak calling was performed using MACS2 (<https://github.com/macs3-project/MACS>) in Signac, resulting in a reproducible, nonoverlapping peak set including covariates. Genes were associated with peaks by identifying nearest gene and TSS to peaks; the significance of the overlap of differential peaks to differential gene expression per cluster was determined using hypergeometric test. To infer TF activity, we performed TF footprinting analysis on the consensus peak-set using TOBIAS (56), as described in our previous study (57). Briefly, this method starts with Tn5 bias correction using the TOBIAS ATACCorrect module, subtracting the background Tn5 insertion cuts highlighting the effect of protein binding. The footprint score was calculated by TOBIAS ScoreBigWig, which measures both accessibility and depth of the local footprint, thus correlating with the presence of a TF at its target loci, and the chromatin accessibility of the regions where this TF binds. To match footprints to potential TF binding sites and to estimate TF binding activity on its target loci, we applied TOBIAS BINDetect module to the corrected ATACseq signals within peaks, with TF motif position weight matrices from TRASFAC Pro. To avoid motif redundancy, we clustered the motifs according to their sequence similarity using TOBIAS ClusterMotifs and chose one motif per TF that is most similar to others in the same cluster. TOBIAS BINDetect compares the positions and activities of TF footprinted sites in disease or control per cluster. Each footprint site was assigned a log2FC between two conditions, representing whether the binding site has larger or smaller TF footprint scores in

comparison. To calculate statistics, a background distribution of footprint scores is built by TOBIAS randomly subsetting peak regions at ~200-bp intervals, and these scores were used to calculate a distribution of background log2FCs for each comparison of two conditions. The global distribution of log2FCs per TF was compared to the background distributions to calculate a differential TF binding score, which represents differential TF activity between two conditions. A *P* value is calculated by TOBIAS by subsampling 100 log2FCs from the background and calculating the significance of the observed change. By comparing the observed log2FC distribution to the background log2FC, the effects of any global differences due to sequencing depth, noise, etc., are controlled. To visualize, we used soGGi to plot TF footprints and bar graphs to show the global TF footprint activity changes comparing ASD versus control.

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SUPPLEMENTARY MATERIALS

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Figs. S1 to S13

Tables S1 to S7

MDAR Reproducibility Checklist

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